

GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

row 22 - INPUT_A_BYTES_PER_ELEMENT

$$\text{Bytes}_{\text{pElement}}^{(\text{matA})} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matA})}]$$

row 23 - INPUT_B_BYTES_PER_ELEMENT

$$\text{Bytes}_{\text{pElement}}^{(\text{matB})} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matB})}]$$

row 25 - OUTPUT_BYTES_PER_ELEMENT_AFTER_DOWN_CONVERSION

$$\text{Bytes}_{\text{pElement}}^{(\text{D}\downarrow)} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matD}\downarrow)}]$$

row 33 - EU_COUNT

$$|\text{EU}| = |\text{XeCore}|_{\text{pXeCU}} \times |\text{EU}|_{\text{pXeCore}} \times |\text{XeCU}|$$

row 102 - TOTAL_L2_SIZE_B_FOR_A_SINGLE_INSTANCE

$$|\text{L2Bytes}| = \text{L2BankSize}_{\text{MB}} \times |\text{L2Banks}|_{\text{pXeCU}} \times 1024 \times 1024$$

row 125 - MAX_POSSIBLE_HBM_BW_FREQ_B_CLK

$$\text{MemBW}_{\text{BpClk}}^{\text{max}} = \frac{\text{MemBW}_{\text{GBps}}^{\text{max}}}{f_{\text{GHz}}^{(\text{GT})}}$$

row 36 - MMA_MAC_THROUGHPUT_PER_XECORE

$$\tau_{\text{mMACpClk} \cdot \text{XeCore}}^{(\text{peak})} = \min\left(\frac{4}{\frac{\text{FLOOR}(\text{Bytes}_{\text{pElement}}^{(\text{matA})} \times 2, 1)}{2}}, \frac{4}{\frac{\text{FLOOR}(\text{Bytes}_{\text{pElement}}^{(\text{matB})} \times 2, 1)}{2}}\right) \times \text{D}_{\text{DPAS}} \times 16 \times |\text{EU}|_{\text{pXeCore}}$$

row 42 - CLKS_PER_DPAS

$$\text{CLKS}_{\text{DPAS}} = \frac{\text{CLKS}_{\text{DPAS}_{\text{num}}}}{\text{CLKS}_{\text{DPAS}_{\text{den}}}}$$

$$\text{CLKS}_{\text{DPAS}_{\text{num}}} = \text{M}_{\text{pThread}} \times \text{K}_{\text{pThread}} \times \text{N}_{\text{pThread}}$$

$$\text{CLKS}_{\text{DPAS}_{\text{den}}} = \frac{\text{CLKS}_{\text{DPAS}_{\text{den}_{\text{num}}}}}{|\text{EU}|_{\text{pXeCore}}}$$

$$\text{CLKS}_{\text{DPAS}_{\text{den}_{\text{num}}}} = \tau_{\text{mMACpClk-XeCore}}^{(\text{peak})}$$

row 13 - WAVES

$$\text{N}_{\text{waves}} = \text{CEILING}(|\text{Tiles}|_{\text{N}}^{(\text{GPU})}, 1) \times \text{CEILING}(|\text{Tiles}|_{\text{M}}^{(\text{GPU})}, 1)$$

row 14 - TG_TILES_IN_N

$$|\text{Tiles}|_{\text{N}}^{(\text{TG})} = \frac{|\text{Tiles}|_{\text{N}_{\text{num}}}^{(\text{TG})}}{|\text{Tiles}|_{\text{N}_{\text{den}}}^{(\text{TG})}}$$

$$|\text{Tiles}|_{\text{N}_{\text{num}}}^{(\text{TG})} = \text{N}_{\text{dim}}$$

$$|\text{Tiles}|_{\text{N}_{\text{den}}}^{(\text{TG})} = \text{W}_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}$$

row 15 - TG_TILES_IN_M

$$|\text{Tiles}|_{\text{M}}^{(\text{TG})} = \frac{\text{M}_{\text{dim}}}{\text{H}_{\text{element}}^{(\text{ThreadGroup})}}$$

row 16 - TG_CLUSTER_TILES_IN_N

$$|\text{Tiles}|_N^{(\text{TG Cluster})} = \frac{|\text{Tiles}|_N^{(\text{TG Cluster})}_{\text{num}}}{W_{\text{element}}^{(\text{XeCoreCluster})}}$$

$$|\text{Tiles}|_N^{(\text{TG Cluster})}_{\text{num}} = N_{\text{dim}}$$

row 17 - TG_CLUSTER_TILES_IN_M

$$|\text{Tiles}|_M^{(\text{TG Cluster})} = \frac{|\text{Tiles}|_M^{(\text{TG Cluster})}_{\text{num}}}{H_{\text{element}}^{(\text{XeCoreCluster})}}$$

$$|\text{Tiles}|_M^{(\text{TG Cluster})}_{\text{num}} = M_{\text{dim}}$$

row 18 - XECU_TILES_IN_N

$$|\text{Tiles}|_N^{(\text{XeCU})} = \frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCUTile})}}$$

row 19 - XECU_TILES_IN_M

$$|\text{Tiles}|_M^{(\text{XeCU})} = \frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCUTile})}}$$

row 20 - GPU_TILES_IN_N

$$|\text{Tiles}|_N^{(\text{GPU})} = \frac{|\text{Tiles}|_N^{(\text{GPU})}_{\text{num}}}{W_{\text{XeCU}}^{(\text{GPUTile})}}$$

$$|\text{Tiles}|_N^{(\text{GPU})}_{\text{num}} = \frac{|\text{Tiles}|_N^{(\text{GPU})}_{\text{num}_{\text{num}}}}{W_{\text{element}}^{(\text{XeCUTile})}}$$

$$|\text{Tiles}|_{N_{\text{num_num}}}^{(\text{GPU})} = N_{\text{dim}}$$

row 21 - GPU_TILES_IN_M

$$|\text{Tiles}|_{M_{\text{num}}}^{(\text{GPU})} = \frac{|\text{Tiles}|_{M_{\text{num}}}^{(\text{GPU})}}{H_{\text{XeCU}}^{(\text{GPUTile})}}$$

$$|\text{Tiles}|_{M_{\text{num}}}^{(\text{GPU})} = \frac{|\text{Tiles}|_{M_{\text{num_num}}}^{(\text{GPU})}}{H_{\text{element}}^{(\text{XeCUTile})}}$$

$$|\text{Tiles}|_{M_{\text{num_num}}}^{(\text{GPU})} = M_{\text{dim}}$$

row 43 - THREAD_WIDTH_IN_UNITS_OF_ELEMENTS

$$W_{\text{element}}^{(\text{Thread})} = N_{\text{pThread}}$$

row 44 - THREAD_HEIGHT_IN_UNITS_OF_ELEMENTS

$$H_{\text{element}}^{(\text{Thread})} = M_{\text{pThread}}$$

row 47 - TG_WIDTH_IN_UNITS_OF_ELEMENT_REALIZED_BY_MULTIPLE_MMA_ITERATIONS

$$W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})} = W_{\text{thread}}^{(\text{ThreadGroup})} \times W_{\text{element}}^{(\text{Thread})}$$

row 48 - TG_HEIGHT_IN_UNITS_OF_ELEMENT

$$H_{\text{element}}^{(\text{ThreadGroup})} = H_{\text{thread}}^{(\text{ThreadGroup})} \times H_{\text{element}}^{(\text{Thread})}$$

row 52 - XECORE_CLUSTER_WIDTH_IN_UNITS_OF_ELEMENT

$$W_{\text{element}}^{(\text{XeCoreCluster})} = W_{\text{TG, keep cluster size as 4}}^{(\text{XeCoreCluster})} \times W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}$$

row 53 - XECORE_CLUSTER_HEIGHT_IN_UNITS_OF_ELEMENT

$$H_{\text{element}}^{(\text{XeCoreCluster})} = H_{\text{TG,keep cluster sizes4}}^{(\text{XeCoreCluster})} \times H_{\text{element}}^{(\text{ThreadGroup})}$$

row 56 - XECU_TILE_WIDTH_IN_UNITS_OF_ELEMENT

$$W_{\text{element}}^{(\text{XeCUTile})} = W_{\text{TG}}^{(\text{XECUTile})} \times W_{\text{element}}^{(\text{ThreadGroup,realized by multiple MMA iterations})}$$

row 57 - XECU_TILE_HEIGHT_IN_UNITS_OF_ELEMENT

$$H_{\text{element}}^{(\text{XeCUTile})} = H_{\text{TG}}^{(\text{XECUTile})} \times H_{\text{element}}^{(\text{ThreadGroup})}$$

row 59 - GPU_TILE_HEIGHT_IN_XECU_UINT

$$H_{\text{XeCU}}^{(\text{GPUTile})} = \frac{|\text{XeCU}|}{W_{\text{XeCU}}^{(\text{GPUTile})}}$$

row 60 - GPU_TILE_WIDTH_IN_UNITS_OF_ELEMETNS

$$W_{\text{element}}^{(\text{GPUTile})} = W_{\text{XeCU}}^{(\text{GPUTile})} \times W_{\text{element}}^{(\text{XeCUTile})}$$

row 61 - GPU_TILE_HEIGHT_IN_UNITS_OF_ELEMETNS

$$H_{\text{element}}^{(\text{GPUTile})} = H_{\text{XeCU}}^{(\text{GPUTile})} \times H_{\text{element}}^{(\text{XeCUTile})}$$

row 63 - MAT_A_INPUT_SIZE_B

$$\text{Size}_B^{(\text{matA})} = M_{\text{dim}} \times K_{\text{dim}} \times \text{Bytes}_{\text{pElement}}^{(\text{matA})}$$

row 64 - MAT_B_INPUT_SIZE_B

$$\text{Size}_B^{(\text{matB})} = K_{\text{dim}} \times N_{\text{dim}} \times \text{Bytes}_{\text{pElement}}^{(\text{matB})}$$

row 65 - MAT_C_INPUT_D_OUTPUT_SIZE_B

$$\text{Size}_B^{(\text{matC,matD})} = M_{\text{dim}} \times N_{\text{dim}} \times \text{Bytes}_{\text{pElement}}^{(\text{D}\downarrow)}$$

row 66 - MAT_D_INTERMEDIATE_SIZE_B

$$\text{Size}_B^{(\text{matD}\downarrow)} = M_{\text{dim}} \times N_{\text{dim}} \times \text{Byte}_{\text{pElement}}^{(\text{matD})}$$

row 103 - WORKING_DATA_SET_SIZE_OF_K_IN_L2_CORRESP_20K_CLOCKS_OF_THREAD_DIVERGENCE

$$|\text{WorkingSet}|_{20\text{Kclksofthreaddivergence}}^{(\text{KinL2})} = \min(20000 \times \frac{K_{\text{pThread}}}{\text{CLKS}_{\text{DPAS}}}, K_{\text{dim}})$$

row 104 - TOTAL_REQUIRED_L2_SIZE_FOR_IDEAL_HIT_RATE_B_FOR_A_SINGLE_INSTANCE_AND_SINGLE_WAVE

$$\begin{aligned} \text{TotalRequiredL2Size}_{\text{matB alwayshit}}^{1 \text{ instance, 1 wave}} &= H_{\text{element}}^{(\text{XeCUTile})} \times \text{Bytes}_{\text{pElement}}^{(\text{matA})} \times |\text{WorkingSet}|_{20\text{Kclksofthreaddivergence}}^{(\text{KinL2})} \\ &+ W_{\text{element}}^{(\text{XeCUTile})} \times \text{Bytes}_{\text{pElement}}^{(\text{matB})} \times |\text{WorkingSet}|_{20\text{Kclksofthreaddivergence}}^{(\text{KinL2})} \\ &+ W_{\text{element}}^{(\text{XeCUTile})} \times H_{\text{element}}^{(\text{XeCUTile})} \times \text{Bytes}_{\text{pElement}}^{(\text{D}\downarrow)} \end{aligned}$$

row 37 - CLK_SPECIFIED_EFFICIENCY

$$T_{\text{clk}}^{(\text{total})} = \frac{T_{\text{clk num}}^{(\text{total})}}{\eta_{\text{systolic}}}$$

$$T_{\text{clk num}}^{(\text{total})} = \frac{T_{\text{clk num num}}^{(\text{total})}}{\tau_{\text{mMACpClk-XeCore}}^{(\text{peak})}}$$

$$T_{\text{clk num num}}^{(\text{total})} = \frac{T_{\text{clk num num num}}^{(\text{total})}}{|\text{XeCU}|}$$

$$T_{\text{clk num num num}}^{(\text{total})} = \frac{T_{\text{clk num num num num}}^{(\text{total})}}{|\text{XeCore}|_{\text{pXeCU}}}$$

$$T_{\text{clk}}^{(\text{total})}_{\text{num}_{\text{num}}_{\text{num}}_{\text{num}}} = M_{\text{dim}} \times K_{\text{dim}} \times N_{\text{dim}}$$

row 69 - TOTAL_L2_READ_B

$$\begin{aligned} \text{L2Rd}_B^{(\text{total})} &= \text{Size}_B^{(\text{matA})} \times \text{CEILING}\left(\frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}}, 1\right) \\ &+ \text{Size}_B^{(\text{matB})} \times \text{CEILING}\left(\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{ThreadGroup})}}, 1\right) \end{aligned}$$

row 70 - TOTAL_L2_WRITE_B

$$\text{L2Wr}_B^{(\text{total})} = \text{Size}_B^{(\text{matC, matD})}$$

row 71 - TOTAL_L1_READ_B

$$\begin{aligned} \text{L1Rd}_B^{(\text{total})} &= \text{Size}_B^{(\text{matA})} \times \text{CEILING}\left(\frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{Thread})}}, 1\right) \\ &+ \text{Size}_B^{(\text{matB})} \times \text{CEILING}\left(\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{Thread})}}, 1\right) \end{aligned}$$

row 72 - TOTAL_L1_WRITE_B

$$\text{L1Wr}_B^{(\text{total})} = \text{Size}_B^{(\text{matC, matD})}$$

row 74 - L2_READ_B_XECORE_CLK

$$\begin{aligned} \text{L2RdB}_{p(\text{Clk} \cdot \text{XeCore})} &= \frac{\text{L2RdB}_{p(\text{Clk} \cdot \text{XeCore})_{\text{num}}}}{T_{\text{clk}}^{(\text{total})}} \\ \text{L2RdB}_{p(\text{Clk} \cdot \text{XeCore})_{\text{num}}} &= \frac{\text{L2RdB}_{p(\text{Clk} \cdot \text{XeCore})_{\text{num}_{\text{num}}}}}{|\text{XeCU}| \times |\text{XeCore}|_{p\text{XeCU}}} \end{aligned}$$

$$L2RdB_{p(Cl k \cdot XeCore)_{num_{num}}} = L2Rd_B^{(total)}$$

row 75 - L2_WRITE_B_XECORE_CLK

$$L2WrB_{p(Cl k \cdot XeCore)} = \frac{L2WrB_{p(Cl k \cdot XeCore)_{num}}}{T_{clk}^{(total)}}$$

$$L2WrB_{p(Cl k \cdot XeCore)_{num}} = \frac{L2WrB_{p(Cl k \cdot XeCore)_{num_{num}}}}{|XeCU| \times |XeCore|_{pXeCU}}$$

$$L2WrB_{p(Cl k \cdot XeCore)_{num_{num}}} = L2Wr_B^{(total)}$$

row 76 - L2_READ_WRITE_B_XECORE_CLK

$$L2RdWrB_{p(Cl k \cdot XeCore)} = L2RdB_{p(Cl k \cdot XeCore)} + L2WrB_{p(Cl k \cdot XeCore)}$$

row 77 - L1_READ_B_XECORE_CLK

$$L1RdB_{p(Cl k \cdot XeCore)} = L1RdB_{p(Cl k \cdot EU)} \times |EU|_{pXeCore}$$

row 78 - L1_WRITE_B_XECORECLK

$$L1WrB_{p(Cl k \cdot XeCore)} = L1WrB_{p(Cl k \cdot EU)} \times |EU|_{pXeCore}$$

row 79 - L1_READ_B_EU_CLK

$$L1RdB_{p(Cl k \cdot EU)} = \frac{L1RdB_{p(Cl k \cdot EU)_{num}}}{T_{clk}^{(total)}}$$

$$L1RdB_{p(Cl k \cdot EU)_{num}} = \frac{L1RdB_{p(Cl k \cdot EU)_{num_{num}}}}{|EU|}$$

$$L1RdB_{p(Clk \cdot EU)_{num_num}} = L1Rd_B^{(total)}$$

row 80 - L1_WRITE_B_EU_CLK

$$L1WrB_{p(Clk \cdot EU)} = \frac{L1WrB_{p(Clk \cdot EU)_{num}}}{T_{clk}^{(total)}}$$

$$L1WrB_{p(Clk \cdot EU)_{num}} = \frac{L1WrB_{p(Clk \cdot EU)_{num_num}}}{|EU|}$$

$$L1WrB_{p(Clk \cdot EU)_{num_num}} = L1Wr_B^{(total)}$$

row 83 - L2_READ_MAX_B_XECORE_CLK

$$L2RdB_{p(Clk \cdot XeCore)}^{(max)} = \frac{L2RdB_{p(Clk \cdot XeCore)_{num}}^{(max)}}{|XeCore|_{pXeCU}}$$

$$L2RdB_{p(Clk \cdot XeCore)_{num}}^{(max)} = |XeCore|_{pXeCU} \times 64$$

row 84 - L2_WRITE_MAX_B_XECORE_CLK

$$L2WrB_{p(Clk \cdot XeCore)}^{(max)} = L2RdB_{p(Clk \cdot XeCore)}^{(max)}$$

row 85 - L2_READ_WRITE_MAX_B_XECORE_CLK

$$L2RdWrB_{p(Clk \cdot XeCore)}^{(max)} = L2WrB_{p(Clk \cdot XeCore)}^{(max)}$$

row 90 - L2_READ_B_XECORE_CLK_PCT

$$\eta_{L2RdB/p(Clk \cdot XeCore)} = \frac{\eta_{L2RdB/p(Clk \cdot XeCore)_{num}}}{L2RdB_{p(Clk \cdot XeCore)}^{(max)}}$$

$$\eta_{L2RdB/p(Cl k \cdot XeCore)}_{num} = L2RdB_{p(Cl k \cdot XeCore)}$$

row 91 - L2_WRITE_B_XECORE_CLK_PCT

$$\eta_{L2WrB/p(Cl k \cdot XeCore)} = \frac{\eta_{L2WrB/p(Cl k \cdot XeCore)}_{num}}{L2WrB_{p(Cl k \cdot XeCore)}^{(max)}}$$

$$\eta_{L2WrB/p(Cl k \cdot XeCore)}_{num} = L2WrB_{p(Cl k \cdot XeCore)}$$

row 92 - L2_READ_WRITE_B_XECORE_CLK_PCT

$$\eta_{L2RdWrB/p(Cl k \cdot XeCore)} = \frac{\eta_{L2RdWrB/p(Cl k \cdot XeCore)}_{num}}{L2RdWrB_{p(Cl k \cdot XeCore)}^{(max)}}$$

$$\eta_{L2RdWrB/p(Cl k \cdot XeCore)}_{num} = L2RdWrB_{p(Cl k \cdot XeCore)}$$

row 93 - L1_READ_B_EU_CLK_PCT

$$\eta_{L1RdB/p(Cl k \cdot EU)} = \frac{\eta_{L1RdB/p(Cl k \cdot EU)}_{num}}{L1RdBW_{BpCl k \cdot EU}^{max}}$$

$$\eta_{L1RdB/p(Cl k \cdot EU)}_{num} = L1RdB_{p(Cl k \cdot EU)}$$

row 94 - L1_WRITE_B_EU_CLK_PCT

$$\eta_{L1WrB/p(Cl k \cdot EU)} = \frac{\eta_{L1WrB/p(Cl k \cdot EU)}_{num}}{L1WrBW_{BpCl k \cdot EU}^{max}}$$

$$\eta_{L1WrB/p(Cl k \cdot EU)}_{num} = L1WrB_{p(Cl k \cdot EU)}$$

row 106 - L2_HIT_RATE_ASSUMED_RANDOM_ACCESS_WITHIN_THE_WORKING_DATA_SET_PCT

$$L2HitRate_{random\ WS\ access} = \min\left(\frac{|L2Bytes|}{TotalRequiredL2Size_{matB\ alwayshit}^{1\ instance,\ 1\ wave}}, 1\right)$$

row 107 - L2_MISS_RATE_PCT

$$\text{L2MissRate} = 1 - \text{L2HitRate}_{\text{random WS access}}$$

row 108 - TOTAL_L2_READ_TRAFFIC_B

$$\text{L2RdB}_{\text{total}} = \text{L2RdB}_{\text{B}}^{(\text{total})}$$

row 110 - PROBABILITY_OF_MATA_HIT_IN_L2_DURING_A_NON_FIRST_WAVE_PCT

$$\begin{aligned} P_{\text{L2Hit,after1stwave}}^{(\text{matA})} &= \text{IF}(\text{Size}_{\text{B}}^{(\text{matA})} \\ &\quad + \text{Size}_{\text{B}}^{(\text{matB})} \\ &\quad + \text{Size}_{\text{B}}^{(\text{matC,matD})} \leq |\text{L2Bytes}|, 1.0, \text{IF}(\text{K}_{\text{dim}} > 2 \times |\text{WorkingSet}|_{20\text{Kclksofthreaddivergence}}^{(\text{KinL2})}, 0.0, 1 \\ &\quad - \frac{\text{K}_{\text{dim}} - |\text{WorkingSet}|_{20\text{Kclksofthreaddivergence}}^{(\text{KinL2})}}{|\text{WorkingSet}|_{20\text{Kclksofthreaddivergence}}^{(\text{KinL2})}})) \end{aligned}$$

row 111 - PROBABILITY_OF_MATB_HIT_IN_L2_DURING_A_NON_FIRST_WAVE_PCT

$$\begin{aligned} P_{\text{L2Hit,after1stwave}}^{(\text{matB})} &= \text{IF}(\text{Size}_{\text{B}}^{(\text{matA})} \\ &\quad + \text{Size}_{\text{B}}^{(\text{matB})} \\ &\quad + \text{Size}_{\text{B}}^{(\text{matC,matD})} \leq |\text{L2Bytes}|, 1.0, 0.0) \end{aligned}$$

row 112 - PROBABILITY_OF_MATA_MISS_IN_L2_DURING_A_NON_FIRST_WAVE_PCT

$$P_{\text{L2Miss,after1stwave}}^{(\text{matA})} = 1 - P_{\text{L2Hit,after1stwave}}^{(\text{matA})}$$

row 113 - PROBABILITY_OF_MATB_MISS_IN_L2_DURING_A_NON_FIRST_WAVE_PCT

$$P_{\text{L2Miss,after1stwave}}^{(\text{matB})} = 1 - P_{\text{L2Hit,after1stwave}}^{(\text{matB})}$$

row 116 - TOTAL_HBM_READ_B_AFTER_A_COMPLETION_OF_A_WAVE_CONSIDER_COLD_CACHE

$$\text{MemRdB}_{\text{coldstart}}^{(\text{wave})} = \frac{\text{MemRdB}_{\text{coldstart_num}}^{(\text{wave})}}{2}$$

$$\begin{aligned} \text{MemRdB}_{\text{coldstart_num}}^{(\text{wave})} &= \text{Size}_B^{(\text{matA})} \\ &+ \text{Size}_B^{(\text{matA})} \times \left(\text{CEILING}\left(\frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCUTile})}}, 1\right) - 1 \right) \times P_{\text{L2Miss,after1stwave}}^{(\text{matA})} \\ &+ \text{Size}_B^{(\text{matB})} \\ &+ \text{Size}_B^{(\text{matB})} \times \left(\text{CEILING}\left(\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCUTile})}}, 1\right) - 1 \right) \times P_{\text{L2Miss,after1stwave}}^{(\text{matB})} \\ &+ \text{MemRdB}_{\text{coldstart_num_aux4}}^{(\text{wave})} \times \text{L2MissRate} \end{aligned}$$

$$\begin{aligned} \text{MemRdB}_{\text{coldstart_num_aux4}}^{(\text{wave})} &= \text{L2RdB}_{\text{total}} \\ &- \text{MemRdB}_{\text{coldstart_num_aux4_aux1}}^{(\text{wave})} \end{aligned}$$

$$\begin{aligned} \text{MemRdB}_{\text{coldstart_num_aux4_aux1}}^{(\text{wave})} &= \text{Size}_B^{(\text{matA})} \\ &+ \text{Size}_B^{(\text{matA})} \times \left(\text{CEILING}\left(\frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCUTile})}}, 1\right) - 1 \right) \times P_{\text{L2Miss,after1stwave}}^{(\text{matA})} \\ &+ \text{Size}_B^{(\text{matB})} \\ &+ \text{Size}_B^{(\text{matB})} \times \left(\text{CEILING}\left(\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCUTile})}}, 1\right) - 1 \right) \times P_{\text{L2Miss,after1stwave}}^{(\text{matB})} \end{aligned}$$

row 117 - TOTAL_HBM_WRITE_B

$$\text{MemWrB}^{(\text{total})} = \text{Size}_B^{(\text{matC}, \text{matD})}$$

row 118 - TOTAL_HBM_B

$$\text{MemRdWrB}^{(\text{total})} = \text{MemWrB}^{(\text{total})} + \text{MemRdB}_{\text{coldstart}}^{(\text{wave})}$$

row 121 - HBM_READ_B_CLK

$$\text{MemRdB}_{\text{pClk}} = \frac{\text{MemRdB}_{\text{coldstart}}^{(\text{wave})}}{T_{\text{clk}}^{(\text{total})}}$$

row 122 - HBM_WRITE_B_CLK

$$\text{MemWrB}_{\text{pClk}} = \frac{\text{MemWrB}^{(\text{total})}}{T_{\text{clk}}^{(\text{total})}}$$

row 123 - HBM_TOTAL_B_CLK

$$\text{MemRdWrB}_{\text{pClk}} = \frac{\text{MemRdWrB}^{(\text{total})}}{T_{\text{clk}}^{(\text{total})}}$$

row 126 - HBM_BW_PCT

$$\eta_{\text{MemBW}} = \frac{\text{MemRdWrB}_{\text{pClk}}}{\text{MemBW}_{\text{BpClk}}^{\text{max}}}$$

row 127 - GTI_READ_BW_PCT

$$\eta_{\text{GTIRdBW}} = \frac{\text{MemRdB}_{\text{pClk}}}{\text{GtiRdBW}_{\text{BpClk}}^{\text{max}}}$$

row 128 - GTI_WRITE_BW_PCT

$$\eta_{\text{GTIWrBW}} = \frac{\text{MemWrB}_{\text{pClk}}}{\text{GtiWrBW}_{\text{BpClk}}^{\text{max}}}$$